

Illustrating the Quantum Anomalous Hall Effect Through the Spin-Orbit Interaction on a Square Lattice

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(Dated: June 6, 2025)

Abstract: The Quantum Anomalous Hall Effect (QAHE) has been well studied in recent years for its connections to topological materials and dissipationless chiral edge states. Excitement about the QAHE was sparked after similar phenomena to the Quantum Hall Effect were observed without an external magnetic field. This has contributed not only to the study of topological phases of matters but also to fields of practical importance such as spintronics, topological quantum computing and high-efficiency electronics. However, to our knowledge, there are no detailed calculations using a tight-binding Hamiltonian of spins on a square lattice with spin-orbit coupling to demonstrate a relatively simple example of the QAHE and a topological insulator. Many efforts seem to be focused on more complicated lattice structures, such as the honeycomb lattice, possibly due to excitement about graphene, but we believe an example on a 2D square lattice with a simple tight-binding Hamiltonian illustrates the phenomena of the QAHE and topological insulators more clearly.

Keywords: Quantum Hall Effect (QHE), Quantum Spin Hall Effect (QSHE), Quantum Anomalous Hall Effect (QAHE), Time Reversal Symmetry (TRS), Spin-Orbit Interaction (SOI), Spin-Orbit Coupling (SOC)

I. INTRODUCTION

The Quantum Hall Effect (QHE) was discovered in 1980 by Klaus von Klitzing when he observed quantized Hall conductance in 2D electron gases (using silicon-based MOSFETs) subjected to perpendicular magnetic fields (Klitzing [1]). The Hall Effect is observed when a voltage and transverse magnetic field are applied to an electrical conductor and a transverse voltage is observed. In two-dimensional electronic systems at low temperature and strong external magnetic fields the Hall resistance (ratio of the Hall voltage to the imposed longitudinal current) is quantized; this is known as the Quantum Hall Effect (QHE). Figure 1 provides a visual summary of the different kinds of Hall Effects known today.

In his original paper, Klitzing used the QHE to calculate the fine structure constant to higher precision using the observed values of the quantized Hall conductance. The study of this new phenomenon fundamentally changed perspectives on new states of matter from being

characterized by symmetry breaking and order parameters to topological invariants, such as the Chern number. However, the QHE requires strong, externally applied magnetic fields and low temperatures. Klitzing used magnetic fields of up to 18 T in his original experiments. Thereafter, the search for materials that could exhibit quantized Hall conductance without large magnetic fields became important.

In 1988, Haldane proposed a 2D lattice model to observe the QHE without external magnetic fields (Haldane [2]), which is now called the Anomalous Quantum Hall Effect (QAHE). Haldane used a tight-binding Hamiltonian with real nearest-neighbor hopping, complex next-nearest-neighbor hopping and a local magnetic flux perpendicular to each unit cell via a periodic vector potential on a honeycomb lattice to study graphite. It was not clear, however, how to devise such a material. In 2006, this was solved by realizing that the spin-orbit interaction (SOI) between the electron and nucleus acts as the system's own internal magnetic field. In the rest frame of the orbiting electron, there is a magnetic field due to the nucleus. Therefore, no externally applied magnetic fields and low temperature environments are needed to observe Quantum Hall-like effects. Now, counter propagating edge states are due to electrons with different spins being pushed in different directions due to the SOI. These are different from the edge states observed in the QHE, which are due to partially occupied Landau levels near the edge that allow electrons to propagate along the edges. The description of these edge states and corresponding band structures leads to the study of topological insulators. 3D topological insulators such as Bi₂Se₃ and Bi₂Te₃ have been shown to exhibit these effects. Additionally, the predicted band structure of Bi₂Se₃ using the SOI has been experimentally visualized using Angle-Resolved Photoemission Spectroscopy (ARPES) and further validates this theory.

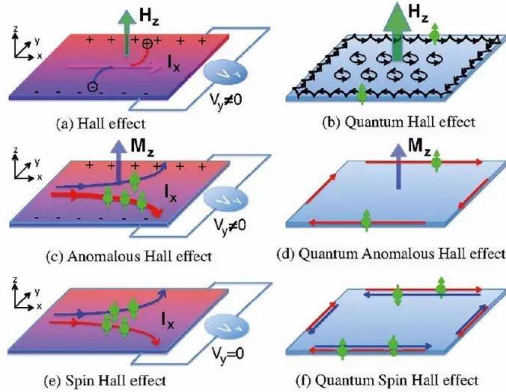


FIG. 1: Different kinds of Hall Effects.

Today, the QAHE is well understood, however, we acknowledge a lack of approachable examples and detailed calculations showing the use of the SOI to demonstrate effects such as chiral edge states and non-trivial topological band structures. In this study, we demonstrate the QAHE in the following steps. First, we recall the band structure of a simple tight-binding Hamiltonian on a lattice of spins, which is notably degenerate due to its spatial symmetry. Then, we observe a splitting of this degeneracy (into a valence and conduction band) by incorporating the SOI through a modified tight-binding Hamiltonian. Finally, we show that this resulting band structure demonstrates the Quantum Spin Hall Effect (QSHE) and not the QAHE until further symmetries are broken by adding extra terms to the Hamiltonian. In thus doing, we understand the fundamental differences between the QSHE and the QAHE: time-reversal symmetry breaking, non-zero topological invariants, and non-zero Hall conductance.

II. DISCUSSION

A. Rotational Symmetry Breaking with Magnetic Fields

The spin-orbit interaction (SOI) is a relativistic effect that couples the particle's spin degree of freedom to the magnetic field it experiences by orbiting the nucleus. The energy of this interaction is given by

$$\Delta H = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad (1)$$

which splits atomic energy levels and we can expect the same to occur in a condensed matter system. The splitting of energy levels due coupling with the magnetic field is physically equivalent to what is observed in the Zeeman effect where spectral lines split due to the presence of a static external magnetic field. The magnetic field in (1) is actually the nuclear electric field, but it is experienced as a magnetic field in the electron's rest frame. In the language of atomic physics, we can see this coupling more explicitly with the addition of angular momentum theory,

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \rightarrow \mathbf{L} \cdot \mathbf{S} = \frac{1}{2} [J^2 - L^2 - S^2], \quad (2)$$

where $L^2 = l(l+1)$ and $S^2 = s(s+1)$ are operators that act on the electron's orbital and spin degrees of freedom (given by the quantum numbers l and s), respectively, and J^2 is an operator that acts on the electron's new degree of freedom (given by the quantum number j) defined by the addition of its spin and orbital-angular momenta. The theory of addition of angular momentum shows that

$$|l - s| \leq j \leq l + s. \quad (3)$$

Since j can take on multiple values for combinations of l and s , the atomic energy levels are split by this new degree of freedom.

B. Incorporating SOC into the Tight-Binding Hamiltonian

We turn our attention to a condensed matter system and make the conjecture that the spin-orbit interaction will split band degeneracies. Consider a 2D square lattice of spins that are allowed to hop between nearest neighbours using a tight-binding Hamiltonian given by

$$\hat{H}_{TB} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + c_{j,\sigma}^\dagger c_{i,\sigma}). \quad (4)$$

where t is the hopping integral from the particle with spin σ' at site j to the particle with spin σ at site i . $c_{i,\sigma}^\dagger$ and $c_{i,\sigma}$ are the creation and annihilation operators, respectively, of spins at these sites. The fundamental assumption of this Hamiltonian is that electrons are tightly bound to their respective atoms and interaction can only take place over a short range. This is the reason for only considering nearest neighbour interactions where the hopping integral t controls the strength of these interactions. We notably do not consider Coulomb interactions between electrons. This is a good approximation for metals where electrons are approximately free or where the electric potential from other electrons is highly screened throughout the lattice. Interparticle interactions are often handled through Density Functional Theory or the Hartree-Fock approximation. In this study, however, we are interested in single particle dynamics.

It is typically useful to analyze the Hamiltonian in momentum space (k -space) by taking its Fourier Transform of the creation and annihilation operators.

$$c_{i,\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{ikx_i} c_{i,k}, \quad (5)$$

$$c_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ikx_i} c_{i,k}^\dagger. \quad (6)$$

Since we are in momentum space under periodic boundary conditions we can traverse the reciprocal lattice by adding or subtracting reciprocal vectors we recast x_j as $x_j = x_i + G$, where G is a reciprocal lattice vector. This shows the very important feature that the tight-binding Hamiltonian has spatial symmetry and leads to a degeneracy in the band structure as later we will see. The Hamiltonian then becomes

$$\hat{H}_{TB} = -\frac{t}{N} \sum_{k,k'} \sum_{x_i, \sigma} c_{k,\sigma}^\dagger R_k^{\sigma, \sigma'} c_{k',\sigma} e^{i(k-k') \cdot x_i} e^{ik' \cdot G} + \text{h.c.}, \quad (7)$$

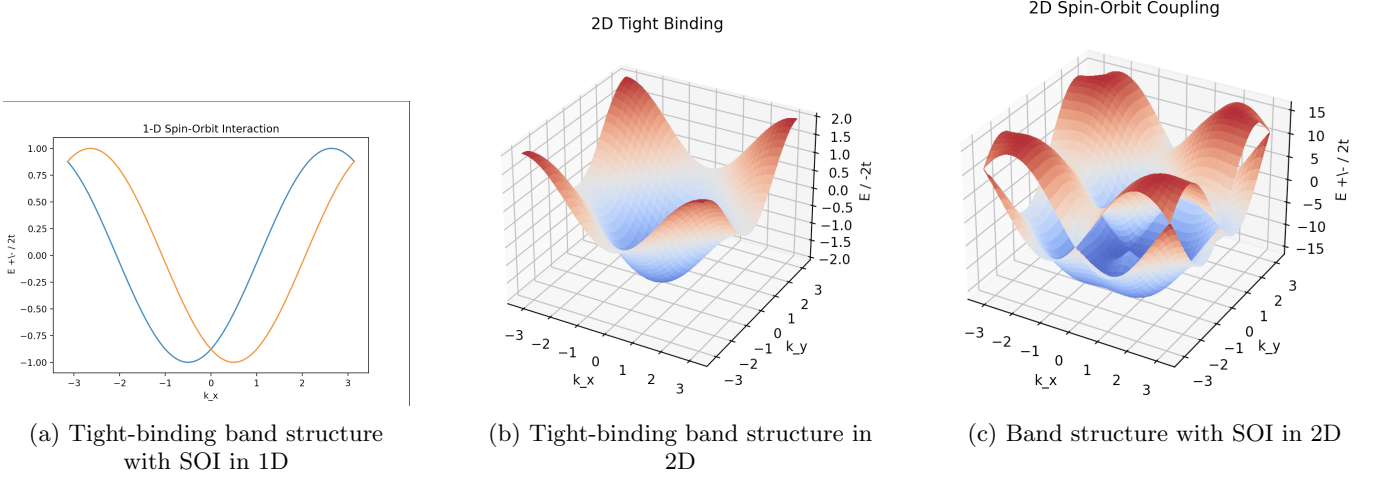


FIG. 2: Tight-binding band structures with and without SOI.

then using the fact that $\sum_{x_i} e^{i(k-k') \cdot x_i} = N\delta_{kk'}$, we have

$$\hat{H}_{TB} = -t \sum_{k,\sigma} c_{k,\sigma}^\dagger R_k^{\sigma,\sigma'} c_{k,\sigma} e^{ik \cdot G} + \text{h.c.}, \quad (8)$$

where $R_k^{\sigma,\sigma'}$ is the Fourier Transform of $R_{i,j}^{\sigma,\sigma'}$. The hermitian conjugate of the first term is denoted as h.c. Now, we can write the Hamiltonian in the following familiar form

$$\hat{H}_{TB} = -t \sum_k \psi_k^\dagger H(\mathbf{k}) \psi_k, \quad (9)$$

where $\psi_k = (c_{k,\sigma}, c_{k,\sigma'})^T$ and

$$H(\mathbf{k}) = \begin{pmatrix} 0 & R_k^{\sigma,\sigma'} e^{ik \cdot G} \\ (R_k^{\sigma,\sigma'})^\dagger e^{-ik \cdot G} & 0 \end{pmatrix}. \quad (10)$$

The energy spectrum is given by $E_\pm = -t (\pm |R_k^{\sigma,\sigma'}|)$.

The eigenvalues are $\pm |R_k^{\sigma,\sigma'}|$. As a quick sanity check, it makes sense that the SOI is on the off-diagonal since it couples different states and breaks degeneracy. Here, $R_{i,j}$ is given by $\exp[i\vec{A} \cdot (\vec{r}_i - \vec{r}_j)]$ (Cole [3]), where $\vec{A}$ is the non-Abelian gauge $(\alpha\sigma_y, \beta\sigma_x, 0)$. The case where $\beta = -\alpha$ corresponds to Rashba spin-orbit coupling. In one dimension $R_{i,j}$ evaluates to

$$\begin{aligned} R_{i,j}^{\hat{x}} &= \exp[\pm i\alpha\sigma_y] = 1 \pm i\alpha\sigma_y - \frac{\alpha^2 \cdot I}{2!} \mp i \frac{\alpha^3 \cdot \sigma_y}{3!} + \dots = \\ &= I \cdot \left(1 - \frac{\alpha^2}{2!} + \frac{\alpha^4}{4!} + \dots\right) \pm i\sigma_y \left(\alpha - \frac{\alpha^3}{3!} + \dots\right) = \\ &= \cos \alpha \pm i \sin \alpha \cdot \sigma_y, \end{aligned} \quad (11)$$

where the $\pm$ denotes the positive and negative x-direction. This is the term that contains the SOI physics in our model. The Hamiltonian can be written in the form

$$H = \mathbf{d}_0 + \mathbf{d}(\vec{k}) \cdot \boldsymbol{\sigma} \quad (12)$$

where $\boldsymbol{\sigma}$ is a vector of Pauli-matrices. These give the eigenvalues (Zhang [4])

$$E_\pm(\mathbf{k}) = \mathbf{d}_0 \pm |\mathbf{d}(\vec{k})|. \quad (13)$$

Since we have already factored our Hamiltonian, which contains all of the physics in the middle 2×2 matrix, we work with $R_{i,j}^{\sigma,\sigma'}$. The Fourier transform is given by

$$\begin{aligned} \mathbf{d}_0 &\xrightarrow{FT} \sum_x \cos \alpha (\cos kx + i \sin kx) = \\ &= \sum_x \cos \alpha \cos kx + \text{Im} \end{aligned} \quad (14)$$

$$\begin{aligned} \mathbf{d}(\vec{k}) &\xrightarrow{FT} \sum_x \pm i \sin \alpha (\cos kx + i \sin kx) = \\ &= \sum_x \mp \sin \alpha \sin kx + \text{Im}, \end{aligned} \quad (15)$$

where Im denotes the imaginary part that is canceled in the sum. The eigenvalues are then given by

$$\begin{aligned}
& \sum_x \cos \alpha \cos kx \pm |\mp \sin \alpha \sin kx| = \\
& \sum_x \cos \alpha \cos kx \pm \sin \alpha \sin kx = \\
& = \sum_x \frac{1}{2} [\cos(kx - \alpha) + \cos(kx + \alpha)] \pm \\
& \quad \pm \frac{1}{2} [\cos(kx - \alpha) - \cos(kx + \alpha)] \\
& \Rightarrow 2 \cos(k \pm \alpha)
\end{aligned} \tag{16}$$

when taking into account hopping in both directions in the First Brillouin Zone (FBZ). The Hamiltonian also contains a $-t$ prefactor, so

$$E_{\pm}(k) = -2t \cos(k \pm \alpha). \tag{17}$$

This is the dispersion produced by the SOI. Without SOI the dispersion for the tight-binding model is $-2t \cos(k)$, which is spatially symmetric and has two degenerate energies corresponding to $G = \pm a$. We can see, therefore, that the effect of the SOI is to split the degeneracy into two distinct bands as seen in figure 2. We notice from the graph that this spatial symmetry is broken, so the energy dispersion is different in the positive direction than in the negative direction. Additionally, adding a Zeeman term would introduce a band gap and make this material an insulator, where the bottom band would be the valence band and the top would be the conduction band. If there is no way to adiabatically transform this material into a normal insulator without first passing through a conduction phase, then it is called a topological insulator.

Now, we add the SOI in the y-direction by adding the analogous hopping matrix to the Hamiltonian

$$R_{i,j}^{\hat{y}} = \cos \beta \pm i \sin \beta \sigma_x. \tag{18}$$

Then, $H(k)$ due to SOI in the y-direction becomes

$$\begin{aligned}
H(k) &= -t[(\cos \beta \mathbb{I} + i \sin \beta \sigma_x) e^{-ik_x a} + \\
& \quad + (\cos \beta \mathbb{I} - i \sin \beta \sigma_x) e^{ik_x a}] = \\
&= -2t[\cos \beta \cos k_x a \mathbb{I} + \sin \beta \sin k_x a \sigma_x]
\end{aligned} \tag{19}$$

In two dimensions the Hamiltonian $H(k)$ in matrix form with the SOI in both positive directions is

$$H(k) = \begin{bmatrix} \cos \alpha + \cos \beta & \sin \alpha + i \sin \beta \\ -\sin \alpha + i \sin \beta & \cos \alpha + \cos \beta \end{bmatrix}. \tag{20}$$

It is important to note that this matrix is written in the spin basis, so the diagonal terms give spin conserving hopping and the off-diagonal terms are spin flips resulting from the SOI. We can decompose this (and any other)

Hamiltonian into a sum of Pauli matrices and the identity, which gives the energy dispersion (Zhang [4])

$$E_{\pm}(\mathbf{k}) = d_0(\mathbf{k}) \pm |\mathbf{d}(\mathbf{k})|, \tag{21}$$

where $\mathbf{d}(\mathbf{k})$ is the vector of coefficients multiplying pauli-matrices and $d_0(\mathbf{k})$ is the coefficient multiplying the identity. Recall that the above hopping matrix was composed of

$$R_{i,j}^{\hat{x}} = \cos \alpha \pm i \sin \alpha \sigma_y \tag{22}$$

$$R_{i,j}^{\hat{y}} = \cos \beta \pm i \sin \beta \sigma_x, \tag{23}$$

which give the decomposition

$$\begin{aligned}
d_0 &= \cos \alpha + \cos \beta \\
\mathbf{d} &= (i \sin \beta, i \sin \alpha, 0).
\end{aligned} \tag{24}$$

Finally, we need to Fourier transform this vector representation of the Hamiltonian to find the dispersion in k -space. Repeating the same calculation from the one dimensional case, α is attached to σ_y , and gives a k_x and β is attached to σ_x and gets k_y since it. After Fourier transforming we obtain

$$\begin{aligned}
d_0(\mathbf{k}) &\xrightarrow{FT} -2t(\cos \alpha \cos k_x + \cos \beta \cos k_y) \\
\mathbf{d}(\mathbf{k}) &\xrightarrow{FT} -2t(\sin \beta \sin k_y, \sin \alpha \sin k_x, 0)
\end{aligned} \tag{25}$$

which gives the energy dispersion in two dimensions

$$E_{\pm}(\mathbf{k}) = d_0(\mathbf{k}) \pm |\mathbf{d}(\mathbf{k})|. \tag{26}$$

We can see that the SOI splits the degenerate energy levels in the normal tight-binding model, which has spatial symmetry, into two distinct bands. The eigenvectors are easy to find with Mathematica:

$$\begin{aligned}
\chi_{\pm}(\mathbf{k}) &= \left(1, \pm \frac{\sin \alpha \sin k_x - \sin \beta \sin k_y}{\sqrt{(\sin \alpha \sin k_x)^2 + (\sin \beta \sin k_y)^2}} \right) = \\
&= (1, \mp i e^{i\phi_{\mathbf{k}}}),
\end{aligned} \tag{27}$$

$$\text{where } \phi_{\mathbf{k}} = \arctan \left(\frac{\sin \beta \sin k_y}{\sin \alpha \sin k_x} \right) = \arctan (d_x/d_y).$$

These eigenvectors are plotted in figure 3 and spin in different directions in k -space if they are in the top (conducting) or bottom (valence) bands. Comparing this to the band structure obtained through tight-binding model without SOI, we can clearly see that the SOI breaks

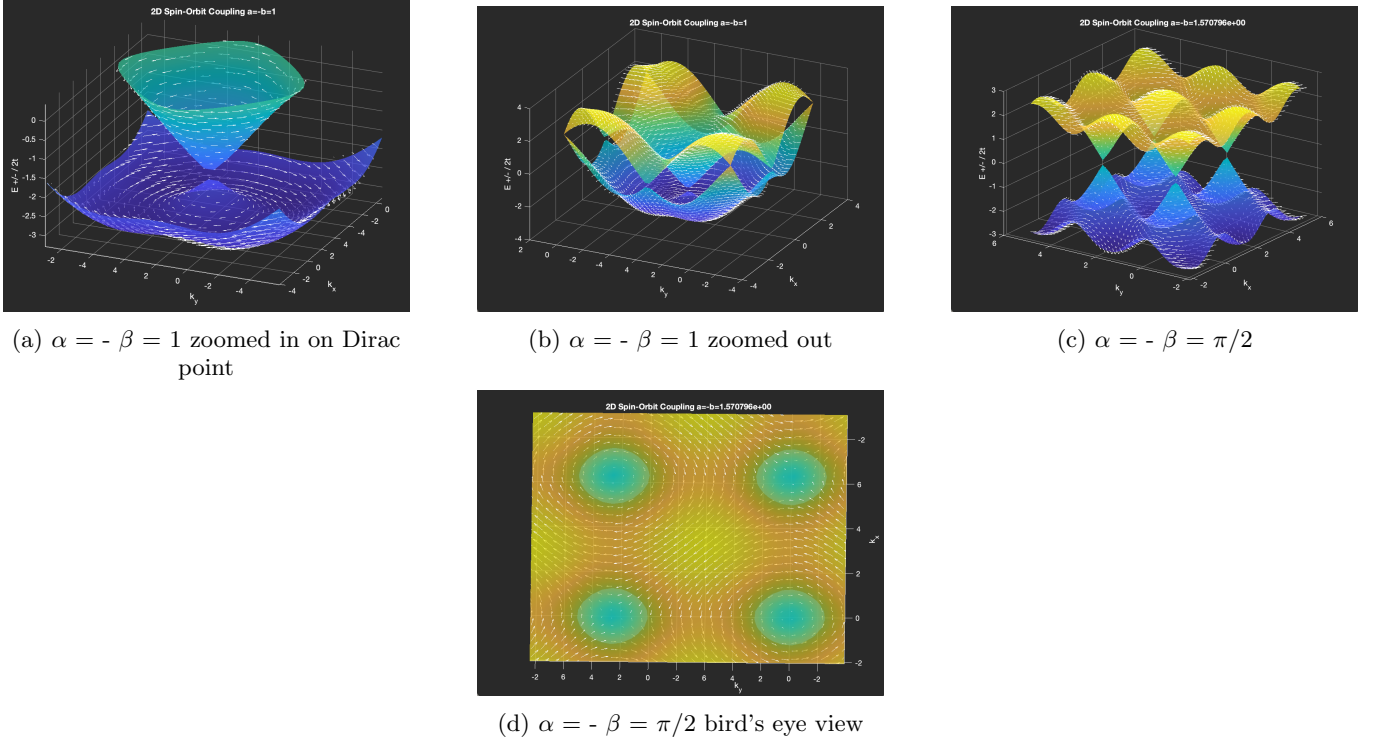


FIG. 3: Band structures with spin textures for different choices of the α and β parameters

the degeneracy and creates a topologically different band structure seen in 3. Finally, the spin textures are shown in figure 3 for Rashba spin-orbit coupling ($\alpha = -\beta$) for the parameters $\alpha = 1$ and $\alpha = \pi/2$, respectively. If we zoom into the middle (figure 3a), we can see the band gap, where the Dirac cone lives, with the spins locked in a circle around the cone.

C. Investigating the Topology of the Band Structure

The conduction and valence bands are gapped and twisted. Notice that the spins are locked in momentum space clockwise in the bottom band and counter clockwise in the upper band. This suggests the band structure of a topological insulator. The next graphs (figures 3b-d) show the same system but for the choice of $\alpha = \pi/2$. A bird's eye view (figure 3d) of this band structure and spin texture also show that this is the band structure of a topological insulator. The spin-momentum locking in the two bands points to non-trivial topology. To determine if the bands are topologically distinct the relevant topological invariant to calculate is the Chern number (C). However, we can easily see that it is zero

$$C = \frac{1}{2\pi} \int_{BZ} d^2k \, \Omega(\mathbf{k}) = \frac{1}{2\pi} \int_{\mathbb{T}^2} d^2k \, \frac{1}{2} \hat{\mathbf{d}} \cdot \left(\frac{\partial \hat{\mathbf{d}}}{\partial k_x} \times \frac{\partial \hat{\mathbf{d}}}{\partial k_y} \right) = 0. \quad (28)$$

As a pedagogical note, the Chern number is the topological invariant that characterizes the global geometry of the band structure and it is the total flux of the Berry curvature $\Omega(\mathbf{k})$ through the first Brillouin zone (Liu *et al.* [5]). To understand this, we first note that the wave-vector $\mathbf{k} \in \mathbb{T}^2$ lives on a 2-Torus since each component $k_i \in S^1$ lives on a circle due to periodic boundary conditions in the reciprocal lattice giving $\mathbf{k} \in S^1 \times S^1 = \mathbb{T}^2$. We can define a mapping for each point in the Brillouin Zone onto the Bloch sphere via the unit vector $\hat{\mathbf{d}} = \frac{\mathbf{E}_\pm(\mathbf{k})}{|\mathbf{E}_\pm(\mathbf{k})|} \in S^2$. The Chern number is then defined through the expression in the second line of equation 28. The interpretation of the Chern number is the amount of (signed) area swept out on the Bloch sphere over the first Brillouin zone (i.e., as $\mathbf{k}$ is varied). The area covered by the unit vector is an integer multiple of the surface area $4\pi n$, where n is the winding number. In other words, the Chern number counts how much $\mathbb{T}^2$ wraps around S^2 . The integrand of the Chern number, $\hat{\mathbf{d}} \cdot \left(\frac{\partial \hat{\mathbf{d}}}{\partial k_x} \times \frac{\partial \hat{\mathbf{d}}}{\partial k_y} \right)$, can be thought of as the Jacobian of the mapping from the 2-torus in momentum space to the Bloch sphere, where $\frac{\partial \hat{\mathbf{d}}}{\partial k_x} \times \frac{\partial \hat{\mathbf{d}}}{\partial k_y}$ is the signed area element and its dot product

with $\hat{\mathbf{d}}$ gives the signed surface area density swept over the sphere. Since the Chern number is calculated to be zero in equation 28 $\mathbb{T}^2$ does not completely cover $\mathbb{S}^2$ and, therefore, is topologically trivial. The consequence of a Chern number of zero is that no QAHE can be identified with this system. The Hall conductivity, $\sigma_H = ne^2/\hbar$, involves the winding number since the Chern number is directly related to the existence of chiral edge modes characteristic of the QAHE (Liu *et al.* [5]). The Chern number, in this context, is called a TKNN invariant and the Hall conductance has been shown to be proportional to the sum of the Chern numbers of each band through the TKNN formula (Thouless *et al.* [6])

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar} \sum_i C_i. \quad (29)$$

This work was, in part, the subject of the 2016 Nobel Prize in Physics shared by Thouless, Haldane and Kosterlitz (The Royal Swedish Academy of Sciences [7]). A full explanation of the TKNN formula and the connection between quantized Hall conductance and the Chern number is beyond the scope of this work but the main idea is to calculate the linear response of the Hall conductance to the perpendicular magnetic field using the Kubo formula. The key result of the TKNN paper is that the Hall conductance using the Kubo formula is quantized by the Chern number and, therefore, given by equation 29. The Chern number has a special significance because, as previously mentioned, it is directly related to the existence of chiral edge modes and a non-zero Hall conductance. More specifically, the Chern number computes the winding number of the mapping from momentum space (a torus) to the space of quantum states (a sphere with radius $|\hat{\mathbf{d}}|$). This is important because it describes how the eigenstates vary across momentum space and, therefore, the topology of the band structure. Recall that the bands are allowed energy values over momentum space. If the topology is non-trivial (i.e., the winding number is non-zero) the wavefunctions twist over the Brillouin zone. At the edges of the material there is a transition to the vacuum, which, of course, has trivial topology ($C = 0$). The wavefunctions need to unwind at the boundary and this corresponds to the chiral edge modes. Therefore, if the Chern number is non-zero, chiral edge modes must exist in order to smoothly interpolate between the non-trivial topology in the bulk ($C \neq 0$) and the vacuum ($C = 0$).

As a result, chiral edge states are topologically protected since the topology of the system is a global property and robust to defects and perturbations. This also means that the chiral edge modes are gapless. The Chern number as a TKNN invariant is valid for insulators (i.e., when the Fermi level lies in the energy gap between the valence and conduction bands). As previously mentioned, the topology of the band structure is robust and invariant under deformations. However, if we do want

to change the topological invariant (i.e., the Chern number) we must close the gap. In this sense, the chiral edge modes are gapless because the gap must be closed at the boundary to change the topological invariant from something non-trivial ($C \neq 0$) in the bulk to the trivial topology ($C = 0$) in the vacuum.

Returning to our example, the simple system of Rashba SOC on a square-lattice does not exhibit the QAHE since the Chern number is zero. However, it is worthwhile to investigate further why we do not see the hallmarks of the QAHE in this system and propose some changes to produce these effects.

First, let us recognize that the Rashba SOC system on a square lattice exhibits the Quantum Spin Hall Effect (QSHE) and not the QAHE. Liu defines the QSHE as a system with SOC in which particles with different spins experiences different forces (Liu *et al.* [5]). This is clearly the case in our system where we observe the spin texture to be different in the valence and conduction bands. This does not automatically imply the existence of helical edge states, which is the hallmark of the QSHE. The Chern number in our simple system is zero, which means the Hall conductance is zero as expected since there are two counter-propagating edge states whose direction of propagation depends on the spin. These are helical edge states whereas chiral edge states refer to unidirectional edge states that do give an overall conductance. Chiral edge states break time-reversal symmetry (TRS) since they are unidirectional whereas counter propagating modes in helical edge states do not. The counter-propagating edge states cancel out so that no Hall conductance is measured, but there is a non-zero spin Hall conductance. To show this, we start by calculating the Berry connection $\Omega(\mathbf{k})$ around the Dirac points using the spinors calculated by equation (27).

$$\Omega_-(\mathbf{k}) = i\langle\chi_-(\mathbf{k})|\nabla_{\phi_k}\chi_-(\mathbf{k})\rangle = -1 \quad (30)$$

Similarly, for the upper-band spinor $\Omega_+(\mathbf{k}) = 1$. The total spin Chern number is then (Sheng [8])

$$C_s = \oint dk [\Omega_+(\mathbf{k}) - \Omega_-(\mathbf{k})] = 2. \quad (31)$$

This gives exactly the expected spin Hall conductance of $\sigma_{SH} = C_s \cdot \frac{e}{4\pi} = \frac{e}{2\pi}$. Therefore, we have shown by calculating a non-zero spin Chern number, which quantifies how much spin-dependent winding there is, that spin-up and spin-down particles flow in opposite directions and lead to spin-filtered edge states. This explains the observation of zero Hall conductance despite the existence of non-trivial topology in the system. This structure is preserved by the inherent TRS in the system since $H(\mathbf{k}) = H(-\mathbf{k})^\dagger$. Given these key aspects (non-zero topological invariant and TRS), the Rashba SOC on a square-lattice system is a kind of gapless topological insulator. To become a true topological insulator and study

the effects of the QAHE, a band gap must be introduced and TRS symmetry broken.

Now, we investigate how the QAHE differs from the QSHE. A key feature of the QSHE, as we have just discovered, is that it does not require time-reversal symmetry (TRS) breaking and has a Chern number of zero. The QAHE, on the other hand, breaks TRS because it has a non-zero Chern number since this implies the existence of chiral edge states and the Hamiltonian is not invariant under the transformation $\mathbf{k} \rightarrow -\mathbf{k}$. Previously we saw that the Chern number in our system is zero because $d_z = 0$. The easiest way to obtain a non-zero Chern number, therefore, is to add a mass dependent TRS-breaking term $m(\mathbf{k})\sigma_z$ to the Hamiltonian (Haldane [2]). If the mass term is just a constant $m(\mathbf{k}) = M$ a uniform band gap is opened but the Chern number remains zero. TRS is broken since this term is proportional to σ_z , but it does not induce any twisting or non-trivial topology. Therefore, in addition to an overall gap opening, we add another term to $m(\mathbf{k})$ that leads to a non-trivial topology, $m(\mathbf{k}) = M - 2t \sin k_x \sin k_y$. The decomposition of the Hamiltonian becomes

$$\begin{aligned} d(\mathbf{k})_x &= -2t \sin \beta \sin k_y \\ d(\mathbf{k})_y &= -2t \sin \alpha \sin k_x \\ d(\mathbf{k})_z &= M - 2t \sin k_x \sin k_y. \end{aligned} \quad (32)$$

We confirm that this term breaks TRS by defining the time-reversal operator $\mathcal{T} = i\sigma_y \mathcal{K}$, where $\mathcal{K}$ is complex conjugation, and checking that $\sigma_x \rightarrow -\sigma_x$, $\sigma_y \rightarrow \sigma_y$ and $\sigma_z \rightarrow -\sigma_z$. Now, we calculate the Chern number around all four Dirac points $(0, 0)$, $(0, \pi)$, $(\pi, 0)$ and (π, π) using the linearization $\sin k_i \approx k_i$ and $\cos k_i \approx 1$ for $k_i \rightarrow 0$. We obtain $C_{k_i} = \frac{1}{2} \text{sgn}(M)$ for each Dirac point. This gives a total Chern number

$$\mathcal{C} = \begin{cases} +2, & \text{if } M > 0 \\ -2, & \text{if } M < 0 \\ 0, & \text{if } M = 0. \end{cases} \quad (33)$$

This leads to a non-zero Hall conductance $\sigma_H = \pm \frac{Me}{2\pi}$. So, we have demonstrated that adding a mass dependent term that breaks TRS leads to a non-zero Chern number, which implies the existence of chiral edge modes creating a net conductance in the absence of an externally applied magnetic field. These features, TRS breaking, SOC and non-zero Chern number show that this system exhibits the QAHE and qualifies as a topological insulator. We summarize the differences between the QAHE and QSHE in table I.

III. CONCLUSION AND OUTLOOK

In this paper, we investigated the effect of spin-orbit coupling on a square lattice of spin-half particles through a tight-binding Hamiltonian with Rashba spin-orbit coupling and showed that it exhibits the Quantum Spin Hall Effect. We learned about the differences between the Quantum Spin Hall effect and the Quantum Anomalous Hall effect, namely that the latter exhibits time-reversal symmetry breaking and has the Chern number as its topological invariant. We also learned about the deep connections between topology and physics through the phenomenon of the topological insulator and identifying the Chern number as a TKNN invariant. Finally, we modified our original Hamiltonian with spin-orbit coupling by adding a mass-dependent term that breaks time-reversal symmetry, induces a non-trivial topology in the band structure and, therefore, produces the Quantum Hall effect and chiral edge modes.

There are many avenues of current research extending the ideas presented in this paper. One example is the study of Moiré superlattices such as twisted MoTe₂ (Abouelkomsan and Fu [9]), which show interaction-driven topological phases. Researchers have also been able to realize the QAHE in certain systems with higher Chern numbers by doping or layering, and creating tunable topological phases (Zhao *et al.* [10]). Deep generative models are also being used to realize new topological materials using symmetry-based analyses (Hong *et al.* [11]).

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Feature	QSHE	QAHE
Time-reversal symmetry (TRS)	Preserved	Broken
Topological invariant	Spin Chern number $C_s \in \mathbb{Z}$	Chern number $\mathcal{C} \in \mathbb{Z}$
Edge states	Helical: opposite spins move in opposite directions	Chiral: unidirectional edge mode
Hall Conductance	0	$\mathcal{C} \cdot \frac{e^2}{h}$
Disorder robustness	Protected by TRS	Topologically protected
Typical materials	HgTe, InAs/GaSb quantum wells	Magnetic topological insulators
External magnetic field required?	No	No

TABLE I: Comparison between the Quantum Spin Hall Effect (QSHE) and the Quantum Anomalous Hall Effect (QAHE).